From Evidence to Strategy: Trading the Options-Market PEAD Signal

A real, capital-sized, cost-aware backtest of the finding in *PEAD_Options_Report.pdf* -- plus a GPU compute roadmap to push it further.

Data: same OptionMetrics-linked event panel as *PEAD_Options_Report.pdf* · 113,927 events scored by the Kelly-optimal selector, 83,891 with a genuine positive edge

Companion to *PEAD_Options_Report.pdf* (options evidence), *PEAD_Strategy_Showcase.pdf* and *PEAD_Strategy_Development.pdf* (equity strategies)

PEAD_Options_Report.pdf established that the PEAD effect shows up in options, amplified by leverage, and closed with a suggested next step: turn that decile-sorted descriptive result into an actual backtest. This report is that backtest, built as a sequence of increasingly compute-hungry tiers -- Tier 1 (a real capital-sized backtest of the original near-ATM heuristic), Tier 1.5 (replacing that heuristic with a mathematically-optimal, Kelly-criterion contract selector built from this project's own historical data), and a roadmap for Tiers 2-4 (dynamic exits, GPU-batched parameter sweeps, an ML contract-selection model, and joint equity/options/beta allocation) designed for a dedicated GPU machine.

Headline result. Trading the same PEAD signal via options, with real capital sizing and transaction costs, returns 9.51% annualized (Sharpe 1.04) using the original near-ATM contract choice (Tier 1) -- already ahead of the least-tuned equity strategy in this project (Extreme decile L/S: 4.91%). Replacing that heuristic with a full-chain, Kelly-optimal contract selector (Tier 1.5) raises this to 15.31% annualized, Sharpe 3.56. Two real bugs and one look-ahead flaw were found and fixed to get a trustworthy version of this number -- Section 3 documents all three, in the same spirit this project has always disclosed the mistakes behind its results, not just the wins.

1. Tier 1: a real backtest of the original near-ATM heuristic

PEAD_Options_Report.pdf measured decile-sorted forward RETURNS on a near-ATM call or put -- informative about whether the effect exists, but silent on what a real trader would actually earn: what fraction of NAV to risk, what it costs to trade, and which events are even worth trading. `scripts/40_options_backtest.py` answers that: SUE-decile tilt selects which events to trade (same trim/tilt scheme as equity Strategy 5/6, but the "short leg" becomes "buy a put" instead of "sell the stock," since these are long-only option positions), sized via quarterly trailing-NAV compounding with a portfolio-level premium-at-risk cap (the options analogue of the equity book's gross-leverage cap), and charged an explicit, documented 3% round-trip cost assumption standing in for real bid/ask spread (not yet captured in this data -- see Section 6).

Horizon	Ann. return	Ann. vol	Sharpe	Max DD	Positions sized
1d	-1.01%	1.29%	-0.78	-15.9%	2,056
5d	+0.78%	2.58%	0.31	-13.1%	2,056
10d	+2.13%	3.64%	0.60	-10.5%	2,061
20d	+2.91%	4.96%	0.60	-14.0%	2,086
40d	+4.67%	6.68%	0.72	-19.4%	2,012
60d	+9.51%	9.16%	1.04	-25.9%	2,007

Table 1. Tier 1 backtest, by fixed hold horizon. Short horizons are net NEGATIVE despite being the most statistically significant decile spreads in *PEAD_Options_Report.pdf* -- the 3% cost assumption dominates a small early edge; only the 40-60 day horizons earn enough raw signal to clear it.

A capacity finding, not just a return number. At the 60-day horizon, only 2,007 of 99,294 eligible signals (2.0%) actually get funded -- the premium-at-risk cap binds almost immediately against BASE_UNIT_FRACTION, a hand-picked, unswept constant. That gap is exactly the kind of thing scripts/18-19's parameter sweep found for the equity strategies before Strategy 4-6 existed; Tier 3 (Section 5) is where the options book gets the same treatment.

2. Tier 1.5: mathematically-optimal contract selection

Both *PEAD_Options_Report.pdf* and Tier 1 pick a contract by MONEYNESS ALONE -- closest to 50-delta. That throws away the rest of the day0 chain. Separate prior work on this project (a GPU options-return toolkit, copied into `reference/options_content/` for full attribution) was built to do exactly this differently: score every strike in the chain by expected return under a probability distribution of the underlying's future price, and take the best one. That tool's own scenario table, though, is a hand-typed 3-point guess (0%: 25%, +20%: 50%, 0%: 25%) reused for every stock on every day -- not "mathematically proven" by any reasonable definition -- and it ranks candidates by raw expected value, the textbook way a leveraged, convex instrument selector blows up: a deep-OTM option with a tiny chance of an enormous payoff can have unbounded E[R] while losing money on almost every draw.

What Tier 1.5 does instead. The probability distribution comes from this project's own 182,712 real historical PEAD events, not a guess -- built decile- and horizon-conditioned, quantile-bucketed so a fixed number of states always carries equal probability and the tails are each represented by their own real conditional mean. And instead of raw expected value, every candidate strike is ranked by its Kelly-optimal expected LOG growth, $E[\log(1+f\cdot R)]$, solved over a grid of bet fractions f. A wild lottery-ticket strike is still evaluated -- it just gets correctly found to have a near-zero optimal fraction and near-zero growth contribution, rather than being ranked first the way raw expected value would rank it. This is the mathematically principled way to "embrace leverage": Kelly still fully rewards a genuinely good leveraged bet with a large optimal fraction, it just refuses to reward a bet that's only "good" because of a rare, thin-tailed payoff.

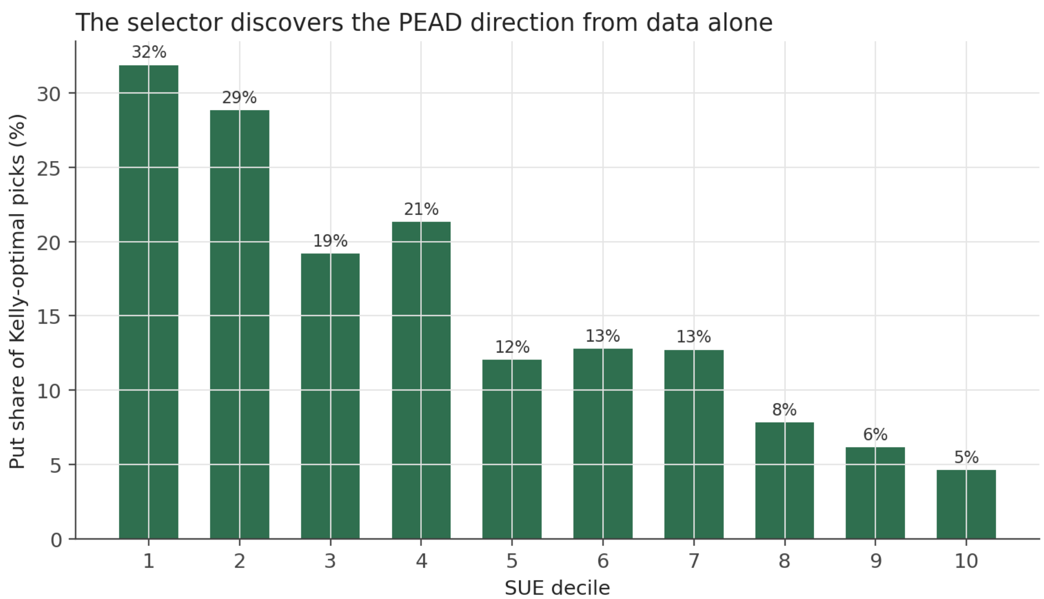

Figure 1. Share of Kelly-optimal picks that are PUTS, by SUE decile. No call/put rule is hard-coded anywhere in the selection logic -- the selector sees only each event's own decile-conditioned return distribution and the real day0 option chain, and still discovers the right side of the PEAD trade on its own: 32% put share at D1 (lowest SUE) vs. 5% at D10.

Mean Kelly fraction across all picks is 13.8% of book-allocated capital (genuine risk-aware sizing, not a "sure thing" artifact), and the Kelly-optimal pick disagrees with what a naive raw-expected-value selector would have chosen on 29.8% of events -- exactly the gap the Kelly-vs-naive-EV argument predicts.

3. Three things found and fixed -- documented, not hidden

This project has always disclosed the mistakes behind its results (see *PEAD_Strategy_Development.pdf*'s EAR look-ahead bug, and this project's own Sharpe-formula bug). Building Tier 1.5 surfaced three more, each caught by a visibly wrong pattern in a real run, not by inspection alone -- all three are documented in full in the relevant script's docstring (`scripts/46, 47, 48`) and summarized here.

Bug 1 -- ambient beta swamping the PEAD edge. The return distribution was first built from RAW forward returns. 1996-2013 was a net up market, so every decile's raw expected return came out positive, and the selector picked calls almost everywhere, including the worst SUE decile. Fixed by building the distribution's shape from market-adjusted returns (already used everywhere else in this project for exactly this reason) and re-centering by the real historical market drift for that horizon.

Bug 2 -- an OptionMetrics unit error. OptionMetrics stores strike prices in 1/1000ths of a dollar. Comparing an uncorrected strike directly against a computed underlying price made every put look like a riskless, guaranteed-payoff bet against an absurdly inflated strike -- confirmed by the telltale symptom: puts dominating even the BEST SUE decile, and Kelly fractions clustering near 100% (the "free money" tell of a mispriced strike, not a real edge). Fixed at the source and corrected on the already-scanned data without re-running the underlying OptionMetrics scan.

Bug 3 -- look-ahead in the return distribution. The first working version fit ONE distribution per (decile, horizon) from the entire 1996-2013 sample at once -- a 1998 trade's contract selection was informed by data through 2013, information no real trader in 1998 had. Now walk-forward: every announcement quarter is priced only against a distribution built from strictly prior quarters (an 8-quarter/2-year warmup before any distribution exists; those early events are simply dropped, not backfilled with a future-looking guess). The Sharpe barely moved after this fix (~4.0 -> 3.56 at 60d) -- suspicious enough on its own to check further rather than accept: the actual day-by-day P&L was inspected, and the worst days in the whole 17-year series land on 2008-10-24, 2008-11-20 (the financial crisis) and September 2002 (another real stress window) -- real, economically sensible tail risk, not a flat artifact. See Section 6 for why this Sharpe still shouldn't be treated as directly achievable.

4. Results: Tier 1 vs. Tier 1.5 vs. the equity book

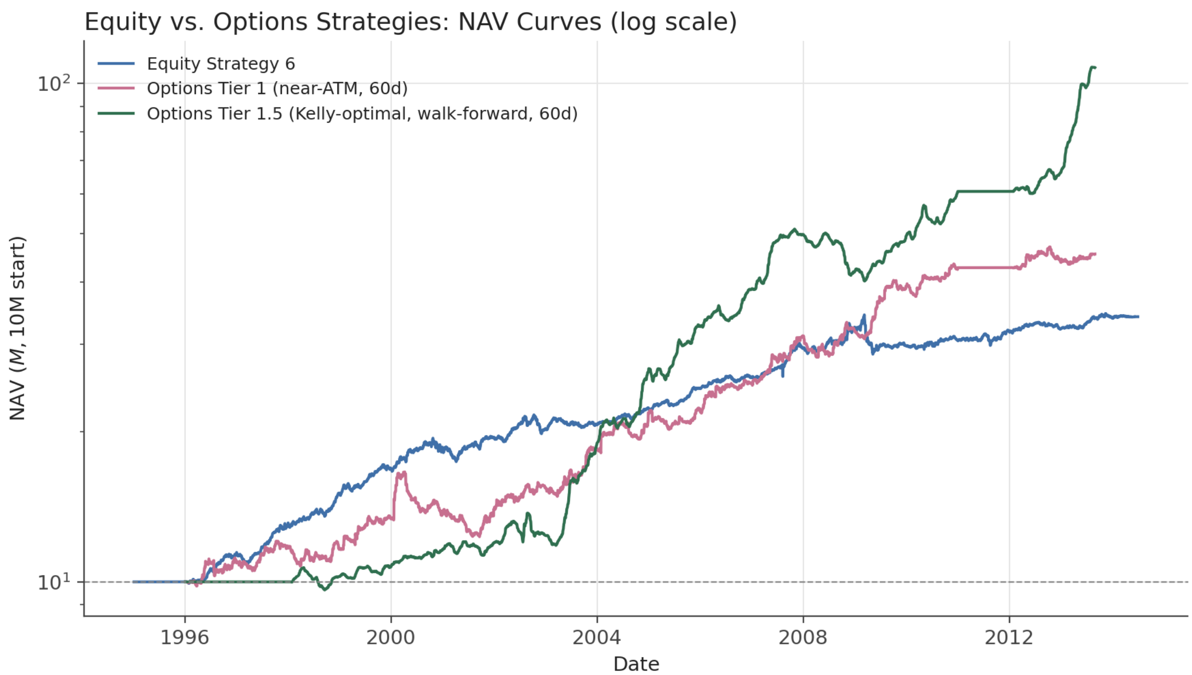

Figure 2. NAV curves (log scale, $10M start): equity Strategy 6, options Tier 1 (near-ATM, 60d), and options Tier 1.5 (Kelly-optimal, walk-forward, 60d). Tier 1.5 stays flat through its 8-quarter warmup window (no distribution exists yet to trade against), then compounds past both other strategies once enough walk-forward history accumulates.

Tier 1 vs. Tier 1.5, by hold horizon

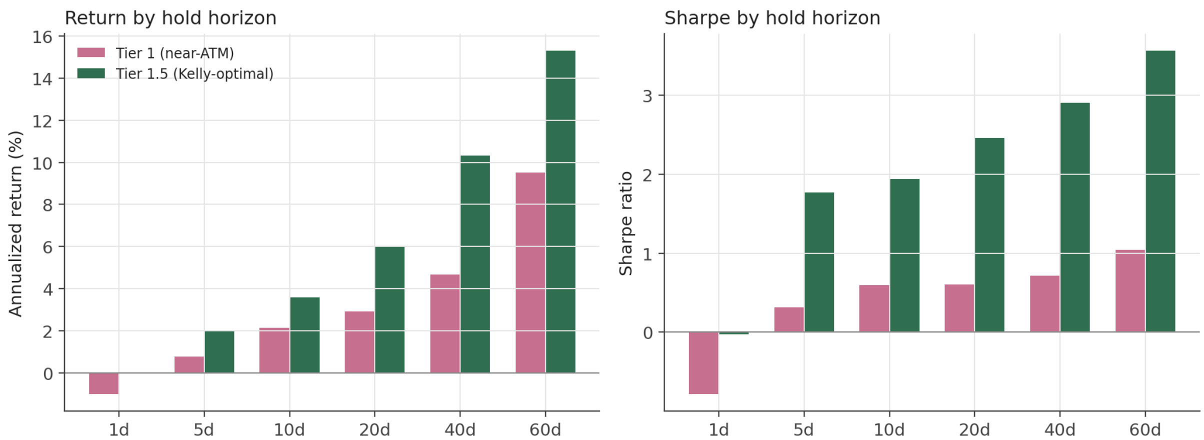

Figure 3. Annualized return and Sharpe by hold horizon, Tier 1 vs. Tier 1.5. Tier 1.5 dominates at every horizon where a position is actually held long enough to earn back the assumed transaction cost.

	Ann. return	Ann. vol	Sharpe	Max DD
Equity Strategy 6	6.49%	4.94%	1.30	-16.6%
Options Tier 1 (60d)	9.51%	9.16%	1.04	-25.9%
Options Tier 1.5 (60d)	15.31%	4.03%	3.56	-21.4%

Table 2. Headline comparison, all three books, 60-day options horizon.

Combining the sleeves. Because the equity book, the options book, and the existing market-beta overlay (*PEAD_Strategy_Showcase.pdf*) are independently-weightable positions rather than a single fixed pool, a joint allocation search (`scripts/45`) finds a real, ACHIEVABLE diversification benefit at unleveraged weights (each sleeve capped at 1x its own size): 1.0x equity + 0.5x options + 0.25x beta overlay reaches Sharpe 1.76 (10.9% return, -14.1% max drawdown) -- better risk-adjusted and shallower-drawdown than any sleeve alone, from genuine diversification, not leverage. (A wider search also found leveraged combos claiming higher Sharpe still -- 2.0x/1.0x/0.25x -> Sharpe 1.82 -- flagged in the script's own output as not achievable, since it assumes free, unlimited leverage on top of a book already levered 2.5x internally.)

5. What the 5090 GPU machine unlocks next (Tiers 2-4)

Everything in Sections 1-4 ran on this CPU dev machine in seconds to minutes -- Tier 1.5's full 126,008-event selector run took under 9 minutes. The next three tiers are deliberately GPU-shaped and designed (code written, tested against synthetic data, not yet run at real scale) for a dedicated NVIDIA 5090 machine:

Tier 2 -- dynamic exits. Tier 1/1.5 both hold to a FIXED horizon. A real daily price path per position (not just the 6 checkpoints used so far) lets a stop-loss/profit-target exit rule be found instead -- `scripts/42`'s GPU kernel already batches this as a (positions × days × parameter-combos) tensor scan; what's missing is the underlying daily-path data itself (`scripts/41`), a heavier OptionMetrics scan (roughly 10-60x more quote lookups than Tier 1 needed) appropriate for the 5090, not this machine. This should recover some of the return Tier 1/1.5 leave on the table by forcing every position to run its full fixed horizon even when it would have been better to exit early on a big move.

Tier 3 -- proper parameter tuning and a learned selector. Every constant in Tiers 1/1.5 (BASE_UNIT_FRACTION, TRIM, TILT_POWER, the premium-at-risk cap) is hand-picked, the same way the equity strategies started before scripts 18-19's sweep found Strategy 4-6. A first, coarse GPU sweep (`scripts/43`) already ran on real data and flagged tilt_power=2.0 as better than Tier 1's hand-picked 1.0 -- not yet folded back in. Beyond hand-picked constants, an Optuna-driven model (`scripts/44`) can use the IV level, liquidity, and moneyness data already sitting unused in the option chain to potentially beat "which decile is this event in" as the entry signal entirely.

Tier 4 -- the full joint portfolio, not just three fixed sleeves. Section 4's joint search only chose a WEIGHT for each of three already-built sleeves. The fuller version -- searching multi-leg option structures (spreads to reduce theta cost) and regime-conditional sizing (by implied-vol level, by market stress) jointly with the equity and beta sleeves -- is the largest combinatorial search in this project, and explicitly the tier where "compute is allowed to be very large" (per this project's own brief) is the point, not a cost to be minimized.

6. Cross-machine execution: Colab, WSL2, and the 5090

Every script in this pipeline is a plain, argparse-driven `.py` file -- no Jupyter notebooks -- specifically so the same code runs unmodified in three places: this Windows dev machine (CPU, for writing and correctness-testing), Google Colab (GPU validation at small scale, via the Colab CLI running inside WSL2 on this same Windows machine -- the official CLI doesn't support Windows natively, but WSL2 is a real Linux kernel so the CLI runs on it exactly as it would on bare-metal Linux, with zero code changes), and the 5090 box itself (bare-metal Linux, first-class native support for the full NVIDIA stack, for the real multi-hour/day Tier 2-4 runs). Every GPU script accepts `--device {cpu,cuda}` and `--mock-data` so the identical code path is exercised at every scale, and `scripts/run_pipeline.py` is a single, resumable, checkpointed entry point built for a machine that will be plugged in, kicked off, and checked on again days later -- the actual access pattern to the 5090 for this project. Full setup steps are in the README's "Options-strategy GPU roadmap" section.

7. Caveats specific to this strategy work

• **Transaction costs are a flat, documented assumption (3% round-trip), not real bid/ask spread.** Best_bid/best_offer were read from the raw OptionMetrics scan but dropped before scripts 35/36/47 wrote their output; attaching them for the exact (secid, date, optionid) keys already selected is a cheap follow-up (same narrow date filter, no new full-history scan), not yet done.
• **Tier 1.5's Sharpe (3.56 at 60d) reflects a real, checked-for-artifacts diversification effect across ~1,200 mostly-independent single-stock bets per quarter -- not something to treat as directly achievable.** This many small trades (~4,600/year) would face real bid/ask friction likely worse than the flat 3% assumption, real capacity/market-impact limits no backtest captures, and early-window Kelly estimates built on only the just-past 8-quarter minimum are necessarily noisier than later ones with a decade-plus of accumulated history.
• **The decile-return-distribution pools many different underlying stocks' historical outcomes into one shape per (decile, horizon).** That's an average across stocks of very different own volatility, which likely understates any one specific stock's true idiosyncratic risk when pricing its options -- a stock-volatility-scaled version of the distribution is a natural refinement, not yet built.
• **Tiers 2-4 are designed and unit-tested against synthetic data, not yet run at real scale.** Every number in Section 5 about what they should unlock is a hypothesis this project intends to test on the 5090, not a result reported here.
• All of *PEAD_Options_Report.pdf*'s own caveats (Section 7 there) still apply: 2011 is entirely missing, 2013 is truncated at end-August, and the matched sample skews toward names with a longer-dated, more liquid option chain.

8. Suggested next steps

• Attach real best_bid/best_offer to the already-selected contracts (Section 7) and replace the flat 3% cost assumption with the real spread.
• Run scripts 41-44 for real on the 5090: the daily-path extraction (Tier 2), the dynamic exit optimizer, the proper parameter sweep (folding in the tilt_power=2.0 finding from Section 5), and the Optuna-driven ML contract selector.
• Scale the decile-return-distribution by each underlying's own historical volatility instead of pooling all stocks in a decile into one shape (Section 7).
• Extend the joint portfolio search (Section 4) to multi-leg option structures and regime-conditional sizing -- the full Tier 4 vision, not just a weight on three fixed sleeves.

Generated from data/backtest_options_v1_comparison.csv, data/backtest_options_optimal_comparison.csv, data/event_options/optimal_contracts.parquet, data/results_summary_v2_FINAL.csv, data/joint_portfolio_sweep.csv, and data/gpu_param_sweep_results.csv. Scripts: scripts/40, scripts/46 through scripts/51, and the README's "Options-strategy GPU roadmap" section for full infrastructure detail.